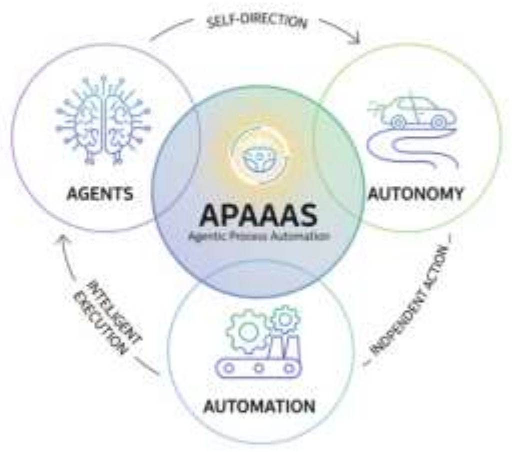
Course Code: R1UC512C

Title: APaaS - Agentic Process Automation as a Service



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Introduction

The course **Agentic Process Automation as a Service (APAAS)** focuses on modern AI systems that automate tasks using intelligent agents. These agents are more advanced than traditional bots because they can reason, use tools, and complete multi-step workflows.

It combines concepts of:

- Artificial Intelligence
- Large Language Models (LLMs)
- Prompt Engineering
- Tool Integration
- Workflow Automation
- Autonomous Decision Making

The course provides both conceptual understanding and practical exposure to real-world AI automation systems.

Objectives of the Internship

The main objectives of APAAS are:

- To understand how AI Agents work in modern automation systems.
- To learn how LLMs are used for reasoning and decision making.
- To study Prompt Engineering techniques.
- To understand Tool Use and Function Calling.
- To understand debugging, testing, and improving agent performance.
- Each objective builds a strong base for working in the field of Agentic AI.

Modules Covered

The course consists of multiple modules.
So far, I have studied:

- Implementing a Basic Agent Structure in Python
- Integrating LLM for Decision Making
- Defining and Integrating Tools
- Simple Input/Output Handling
- Testing and Debugging Agents
- Advanced Prompt Engineering Techniques
- Function Calling and Tool Use

Further advanced modules will be continued later.

Architecture Overview

The course introduced how an AI Agent works internally.

Main Components:

Goal: Task assigned by user

Observation: Current information available

Think: LLM decides next step

Action: Calls tool/function

Memory: Stores history

Agent Flow:

User Request → Think → Use Tool → Observe Result → Final Output

This is the foundation of modern AI automation systems.

Prompt Engineering & LLM Integration

Large Language Models act as the brain of the agent.
Examples: GPT, Claude, Gemini
Prompt engineering teaches how to give effective instructions to LLMs.

Topics Learned:

- Role prompting
- Structured prompting
- Tool prompts
- Iterative prompt refinement
- Testing prompts for better results

Good prompts improve reasoning accuracy and output quality.

Function Calling and Tool Use

LLMs cannot directly access live systems.
Therefore tools are connected.

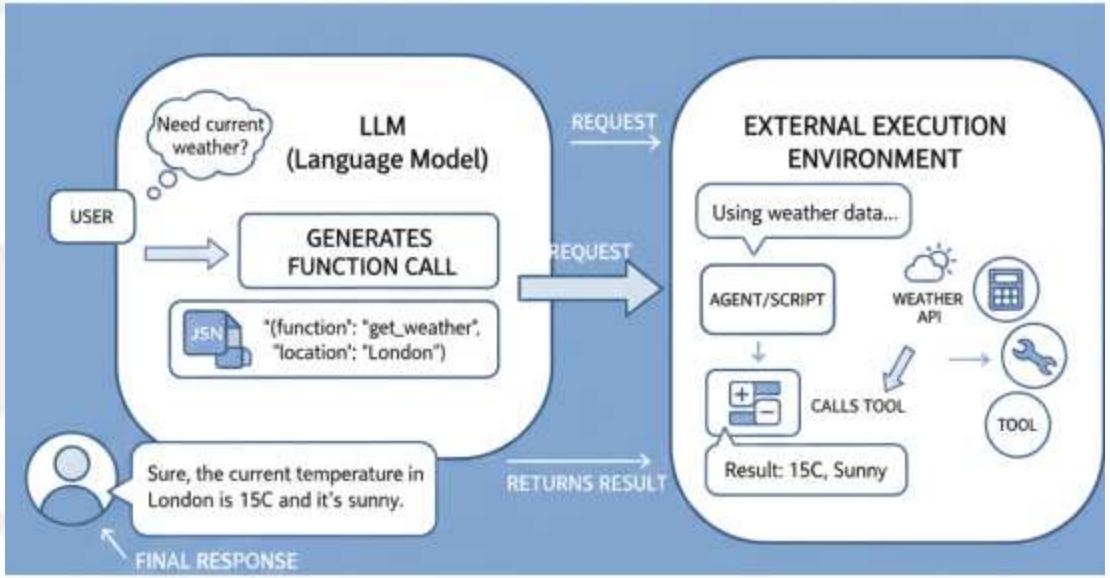
Example Tools:

Calculator
Search Engine
Weather API
Database Query
Email Sender

Working Process:

User asks task → LLM chooses tool → Tool executes → Result returns → Final response generated.

This makes AI agents useful in real-world automation.



Results and Key Learnings

Knowledge Gained:

- How AI agents are structured
- How LLMs make decisions
- How prompts control model behavior
- How tools extend LLM capabilities

Skills Developed:

- Logical thinking
- Workflow understanding
- AI system design concepts
- Debugging complex flows
- Practical understanding of agentic AI

Conclusion and Future Work

Conclusion:
APAAS introduced me to the future of intelligent automation using AI Agents. It helped me understand how autonomous systems are designed using LLMs, prompts, tools.

Future Work:

- Multi-Agent Systems
- LangGraph
- Real-time AI Automation
- Memory-based Agents
- Advanced Deployment Projects

THANK YOU